

Intermediate Macro

Chapter 14: Financial Markets and Expectations

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Summer 2022



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14.4 Risk, Bubbles, Fads, and Asset Prices



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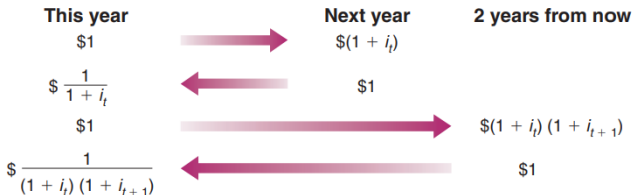
- The **expected present discounted value** of a sequence of payments is the value today (i.e., in current nominal or real units) of the expected sequence of payments.
- The relationship between yield and maturity is introduced via the **yield curve**.
- The concept of **asset bubbles** and the impact on the economy is introduced.

14.1 Expected Present Discounted Values

- To understand why present discounted values are important, consider the problem facing a manager who is deciding whether or not to buy a new machine.
- Once the manager has computed the **expected present discounted value** of the sequence of profits, her problem becomes simpler; she compares two numbers, the expected present discounted value and the initial cost.
- Denote the one-year nominal interest rate by i_t .
- One dollar this year is worth $1 + i_t$ dollars next year.
- Turn the argument around and ask: How much is one dollar *next year* worth this year?
- We say that $\frac{1}{1 + i_t}$ is the *present discounted value* of one dollar next year.

14.1 Expected Present Discounted Values

- Note that the future values are discounted with **discount factor**.
- The rate at which you discount is sometimes called the **discount rate**.



14.1 Expected Present Discounted Values

- $$\$V_t = \$z_t + \frac{1}{(1+i_t)}\$z_{t+1} + \frac{1}{(1+i_t)(1+i_{t+1})}\$z_{t+2} + \dots$$

- $$\begin{aligned} \$V_t = & \$z_t + \frac{1}{(1+i_t)} \$z_{t+1}^e + \frac{1}{(1+i_t)(1+i_{t+1}^e)} \$z_{t+2}^e + \dots \end{aligned} \quad (14.1)$$

Constant Interest Rates and Payments

14.1 Expected Present Discounted Values

- To focus on the effects of the sequence of payments on the present value, assume that interest rates are expected to be constant over time, so that $i_t = i_{t+1}^e = \dots$, and denote their common value by i .
- The present value formula becomes:

$$\$V_t = \$z_t + \frac{1}{(1+i)} \$z_{t+1}^e + \frac{1}{(1+i)^2} \$z_{t+2}^e + \dots \quad (14.2)$$

- In some cases, the sequence of payments for which we want to compute the present value is simple.
- Consider a sequence of equal payments over n year, including this year. In this case, the present value formula in equation (14.2) simplifies to

$$\$V_t = \$z \left[1 + \frac{1}{(1+i)} + \dots + \frac{1}{(1+i)^{n-1}} \right]$$

14.1 Expected Present Discounted Values

- $$\begin{aligned} \$V_t &= \frac{1}{(1+i)}\$z + \frac{1}{(1+i)^2}\$z + \dots \\ &= \frac{1}{(1+i)} \left[1 + \frac{1}{(1+i)} + \dots \right] \$z \end{aligned}$$

14.1 Expected Present Discounted Values

- $$V_t = \frac{1}{1+i} \frac{1}{1 - [1/(1+i)]} z$$

- $$\$V_t = \frac{\$z}{i}$$

Maturity and Risk

14.2 Bond Prices and Bond Yields

- Bonds differ in two basic dimensions:
 - ① Maturity: The **maturity** of a bond is the length of time over which the bond promises to make payments to the holder of the bond.
 - ② Risk: This may be **default risk**, the risk that the issuer of the bond will not pay back the full amount promised by the bond. Or it may be **price risk**, the uncertainty about the price at which you can sell the bond in the future if you want to sell it before maturity.
- Both maturity and risk matter in the determination of interest rates.

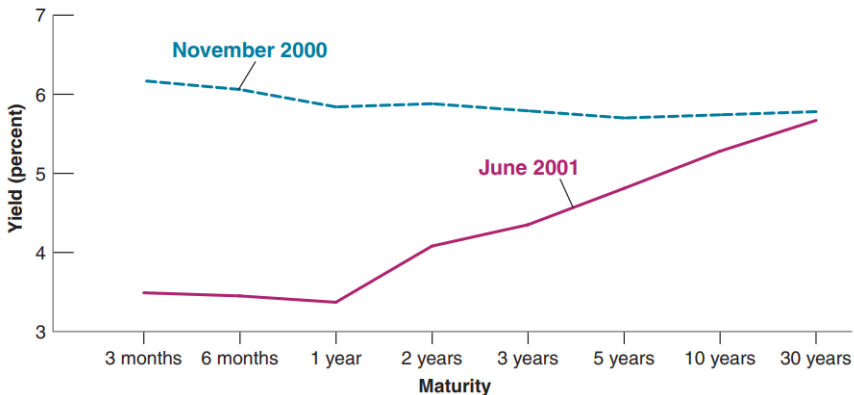
Some Definitions

14.2 Bond Prices and Bond Yields

- Bonds of different maturities each have a price and an associated interest rate called the **yield to maturity**, or simply the **yield**.
- Yields on bonds with a short maturity, typically a year or less, are called **short-term interest rates**.
- Yields on bonds with a longer maturity are called **long-term interest rates**.
- On any given day, we observe the yields on bonds of different maturities, and so we can trace graphically how the yield depends on the maturity of a bond.
- This relation between maturity and yield is called the **yield curve**, or the **term structure of interest rates**.

US Yield Curves: November 1, 2000 and June 1, 2001

14.2 Bond Prices and Bond Yields



Bond Prices as Present Values

14.2 Bond Prices and Bond Yields

- Assume that we have two types of bonds: a bond that promises one payment of \$100 in one year—a one-year bond—and a bond that promises one payment of \$100 in two years—a two-year bond.
- We can price two bonds:

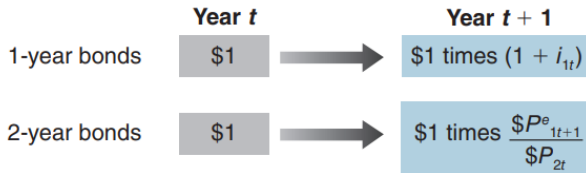
$$\$P_{1t} = \frac{\$100}{1 + i_{1t}} \quad (14.4)$$

$$\$P_{2t} = \frac{\$100}{(1 + i_{1t})(1 + i_{1t+1}^e)} \quad (14.5)$$

An Alternative Derivation

14.2 Bond Prices and Bond Yields

- Let us look at an alternative derivation of equation (14.5).
- This alternative derivation will introduce you to the important concept of **arbitrage**.
- Suppose you have the choice between holding one-year bonds or two-year bonds and what you care about is how much you will have *one year from today*. Which bonds should you hold?



Arbitrage Implication

14.2 Bond Prices and Bond Yields

- If the two bonds offer the same expected one-year return, it follows that

$$1 + i_{1t} = \frac{\$P_{1t+1}^e}{\$P_{2t}} \quad (14.6)$$

- Rewrite equation (14.6) as

$$\$P_{2t} = \frac{\$P_{1t+1}^e}{1 + i_{1t}} \quad (14.7)$$

- Arbitrage implies that the price of a two-year bond today is the present value of the expected price of the bond next year.
- This naturally raises the next question: What does the expected price of one-year bonds next year depend on?

Bond Prices

14.2 Bond Prices and Bond Yields

- Just as the price of a one-year bond this year depends on this year's one-year interest rate, the price of a one-year bond next year will depend on the one-year interest rate next year.

$$\$P_{1t+1}^e = \frac{100}{1 + i_{1t+1}^e} \quad (14.8)$$

- Replacing $\$P_{1t+1}^e$ from equation (14.8) in equation (14.7) gives

$$\$P_{2t} = \frac{100}{(1 + i_{1t})(1 + i_{1t+1}^e)} \quad (14.9)$$

From Bond Prices to Bond Yields

14.2 Bond Prices and Bond Yields

- Bond yields contain the same information about future expected interest rates as bond prices.
- Recall that the yield to maturity on an n -year bond, or equivalently, **n -year interest rate**, is defined as the constant annual interest rate that makes the bond price today equal to the present value of future payments on the bond.
- Take, for example, the two-year bond we introduced previously.

$$\$P_{2t} = \frac{\$100}{(1 + i_{2t})^2} \quad (14.10)$$

- Suppose the bond sells for \$90 today. Then the two-year interest rate i_{2t} is given by $\sqrt{100/90} - 1$, or 5.4%.

Relation of i_{2t} to i_{1t} and i_{1t+1}^e

14.2 Bond Prices and Bond Yields

- Look at equation (14.10) and equation (14.9) and eliminate $\$P_{2t}$ between the two gives

$$\frac{\$100}{(1 + i_{2t})^2} = \frac{\$100}{(1 + i_{1t})(1 + i_{1t+1}^e)}$$

- Rearranging,

$$(1 + i_{2t})^2 = (1 + i_{1t})(1 + i_{1t+1}^e)$$

- A useful approximation to this relation is given by

$$i_{2t} \approx \frac{1}{2}(i_{1t} + i_{1t+1}^e) \quad (14.11)$$

- We can see that long-term interest rates reflect current and future expected short-term interest rates.

Reintroducing Risk

14.2 Bond Prices and Bond Yields

- We have assumed so far that investors did not care about risk.
- Go back to the choice between holding a one-year bond for one year or holding a two-year bond for one year.
- The first option is riskless. The second is risky as you do not know the price at which you will sell the bond in a year.
- You are thus likely to ask for a risk premium to hold the two-year bond, and the arbitrage equation takes the form:

$$1 + i_{1t} + x = \frac{\$P_{1t+1}^e}{\$P_{2t}}$$

- Reorganizing gives:

$$\$P_{2t} = \frac{\$P_{1t+1}^e}{1 + i_{1t} + x}$$

Bond Prices Reflecting the Risk Premium

14.2 Bond Prices and Bond Yields

- The expected price of a one-year bond next year is still given by equation (14.8). So replacing in the previous equation gives:

$$P_{2t} = \frac{\$100}{(1 + i_{1t})(1 + i_{1t+1}^e + x)} \quad (14.12)$$

- Now, to go from prices to yields, let us go through the same steps as before.

$$\begin{aligned} \frac{\$100}{(1 + i_{2t})^2} &= \frac{\$100}{(1 + i_{1t})(1 + i_{1t+1}^e + x)} \\ (1 + i_{2t})^2 &= (1 + i_{1t})(1 + i_{1t+1}^e + x) \end{aligned}$$

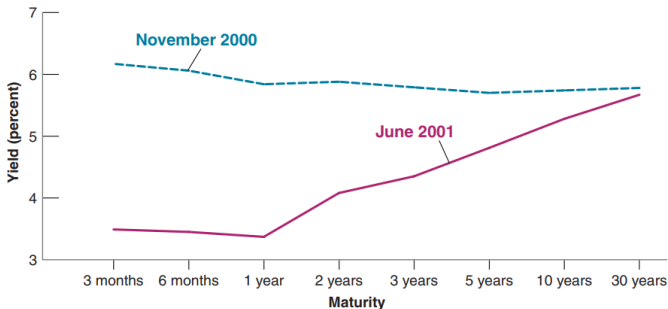
- Using the same approximation gives:

$$i_{2t} = \frac{1}{2}(i_{1t} + i_{1t+1}^e + x) \quad (14.13)$$

Revisiting the Yield Curve

14.2 Bond Prices and Bond Yields

- As the price risk increases with the maturity of the bonds, the risk premium typically increases with maturity.
- This implies that, on average, the yield curve is slightly upward sloping, reflecting the higher risk involved in holding longer maturity bonds.



Interpreting the Yield Curve

14.2 Bond Prices and Bond Yields

- Consider the yield curve for November 1, 2000.
- The fact that the yield curve was downward sloping, something relatively rare, tells us that investors expected interest rates to go down slightly over time, with the expected decrease in rates more than compensating for a rising term premium.
- If we look at the macroeconomic situation at that time, they had good reasons to hold this view.
- At the end of November 2000, the US economy was slowing down.
- Investors expected what they called a *smooth landing*.

Four Ways of Financing of Firms

14.3 The Stock Market and Movements in Stock Prices

- Firms finance in four ways.
 - ① First, they rely on **internal finance**—that is, they use some of their own profits;
 - ② Second, through bank loans—and this is the main channel of **external finance** for small firms;
 - ③ Third, through **debt finance**—bonds;
 - ④ Fourth, through **equity finance**, issuing **stocks**—or **shares**.
- Instead of paying predetermined amounts as bonds do, stocks pay **dividends** in an amount decided by the firm.
- Our focus in this section is on the determination of stock prices.

Stock Prices as Present Values

14.3 The Stock Market and Movements in Stock Prices

- Stock prices, much like bonds, should be equal to the present value of all future cash flows.
- The future cash flows are dividends and a future selling price.
- Let us derive stock prices by looking at the implications of arbitrage between one-year bonds and stocks.
- Suppose you face the choice of investing either in one-year bonds or in stocks for a year. What should you choose?



Stock Prices with Equity Premium

14.3 The Stock Market and Movements in Stock Prices

- In the case of stocks, the risk premium is called the **equity premium**
- Equilibrium requires that the expected rate of return from holding stocks for one year be the same as the rate of return on one-year bonds plus the equity premium:

$$\frac{\$D_{t+1}^e + \$Q_{t+1}^e}{\$Q_t} = 1 + i_{1t} + x$$

where x denotes the equity premium.

- Rewrite this equation as

$$\$Q_t = \frac{\$D_{t+1}^e}{(1 + i_{1t} + x)} + \frac{\$Q_{t+1}^e}{(1 + i_{1t} + x)} \quad (14.14)$$

Further Derivation for Expected Stock Prices

14.3 The Stock Market and Movements in Stock Prices

- The next step is to think about what determines $\$Q_{t+1}^e$.

$$\$Q_{t+1}^e = \frac{\$D_{t+2}^e}{(1 + i_{1t+1}^e + x)} + \frac{\$Q_{t+2}^e}{(1 + i_{1t+1}^e + x)}$$

- Replacing the expected price $\$Q_{t+1}^e$ in equation (14.14) gives

$$\begin{aligned} \$Q_t = & \frac{\$D_{t+1}^e}{(1 + i_{1t} + x)} + \frac{\$D_{t+2}^e}{(1 + i_{1t} + x)(1 + i_{1t+1}^e + x)} \\ & + \frac{\$Q_{t+2}^e}{(1 + i_{1t} + x)(1 + i_{1t+1}^e + x)} \end{aligned}$$

Further Derivation for Expected Stock Prices

14.3 The Stock Market and Movements in Stock Prices

- If we replace the expected price recursively and so on for n years, we get

$$\begin{aligned}
 \$Q_t = & \frac{\$D_{t+1}^e}{(1 + i_{1t} + x)} + \frac{\$D_{t+2}^e}{(1 + i_{1t} + x)(1 + i_{1t+1}^e + x)} + \dots \\
 & + \frac{\$D_{t+n}^e}{(1 + i_{1t} + x) \cdots (1 + i_{1t+n-1}^e + x)} \\
 & + \frac{\$Q_{t+n}^e}{(1 + i_{1t} + x) \cdots (1 + i_{1t+n-1}^e + x)} \quad (14.15)
 \end{aligned}$$

- Look at the last term in equation (14.15), the present value of the expected price in n years. As long as people do not expect the stock price to explode in the future, then as we keep replacing Q_{t+n}^e and n increases, this term will go to zero.

Further Derivation for Expected Stock Prices

14.3 The Stock Market and Movements in Stock Prices

- To see why, suppose the interest rate is constant and equal to i . The last term becomes

$$\frac{\$Q_{t+n}^e}{(1 + i_{1t} + x) \cdots (1 + i_{1t+n-1} + x)} = \frac{\$Q_{t+n}^e}{(1 + i + x)^n}$$

- Suppose further that people expect the price of the stock to converge to some value, call it $\bar{Q}$, in the far future. Then, the last term becomes

$$\frac{\$Q_{t+n}^e}{(1 + i + x)^n} = \frac{\bar{Q}}{(1 + i + x)^n}$$

Real Stock Prices

14.3 The Stock Market and Movements in Stock Prices

- Equation (14.15) reduces to

$$\begin{aligned} \$Q_t = & \frac{\$D_{t+1}^e}{(1 + i_{1t} + x)} + \frac{\$D_{t+2}^e}{(1 + i_{1t} + x)(1 + i_{1t+1}^e + x)} + \dots \\ & + \frac{\$D_{t+n}^e}{(1 + i_{1t} + x) \dots (1 + i_{1t+n-1}^e + x)} + \dots \quad (14.16) \end{aligned}$$

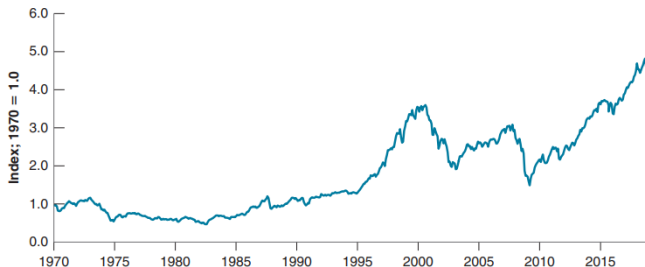
- Equation (14.16) gives the stock price as the present value of *nominal* dividends, discounted by *nominal* interest rates.
- We can rewrite the equation to express the *real* stock price as the present value of *real* dividends, discounted by *real* interest rates.

$$Q_t = \frac{D_{t+1}^e}{(1 + r_{1t} + x)} + \frac{D_{t+2}^e}{(1 + r_{1t} + x)(1 + r_{1t+1}^e + x)} + \dots \quad (14.17)$$

The Stock Market and Economic Activity

14.3 The Stock Market and Movements in Stock Prices

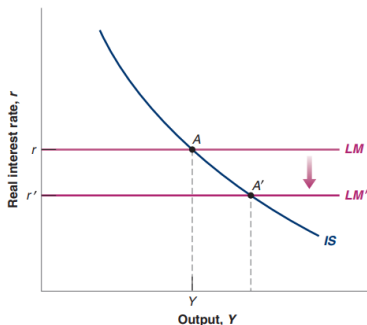
- The following figure shows the large movements in stock prices over the last two decades.
- These movements follow a **random walk** and thus, are unpredictable.



A Monetary Expansion and the Stock Market

14.3 The Stock Market and Movements in Stock Prices

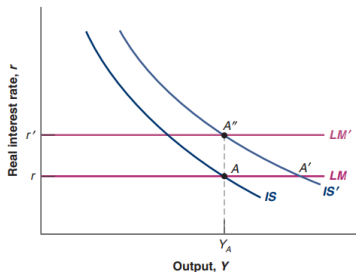
- Suppose the economy is in a recession and the Fed decides to decrease the policy rate.
- How will the stock market react?
- The answer depends on what participants in the stock market expected monetary policy to be before the Fed's move.



An Increase in Consumer Spending and the Stock Market

14.3 The Stock Market and Movements in Stock Prices

- Now consider an unexpected shift of the IS curve to the right, resulting, for example, from stronger-than-expected consumer spending.
- Will stock price go up?
- You might be tempted to say yes, but this answer is not necessarily right.
- The reason is that it ignores the response of the Fed.



Summary

14.3 The Stock Market and Movements in Stock Prices

- Stock prices depend on current and future movements in activity.
- But this does not imply any simple relation between stock prices and output.
- How stock prices respond to a change in output depends on
 - ① what the market expected in the first place,
 - ② the source of the shocks behind the change in output, and
 - ③ how the market expects the central bank to react to the output change.

Stock Prices and Risk

14.4 Risk, Bubbles, Fads, and Asset Prices

- Do all movements in stock and other asset prices come from news about future dividends or interest rate?
- The answer is no, for two reasons:
 - ① Variation over time in perceptions of risk.
 - ② Deviations of prices from fundamental value, namely bubbles or fads.
- In the previous section, we assumed that the equity premium x is constant, but it is not.
- Much of the movement in stock prices comes not just from expectations of future dividends and interest rates but also from shifts in the equity premium.

14.4 Risk, Bubbles, Fads, and Asset Prices

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